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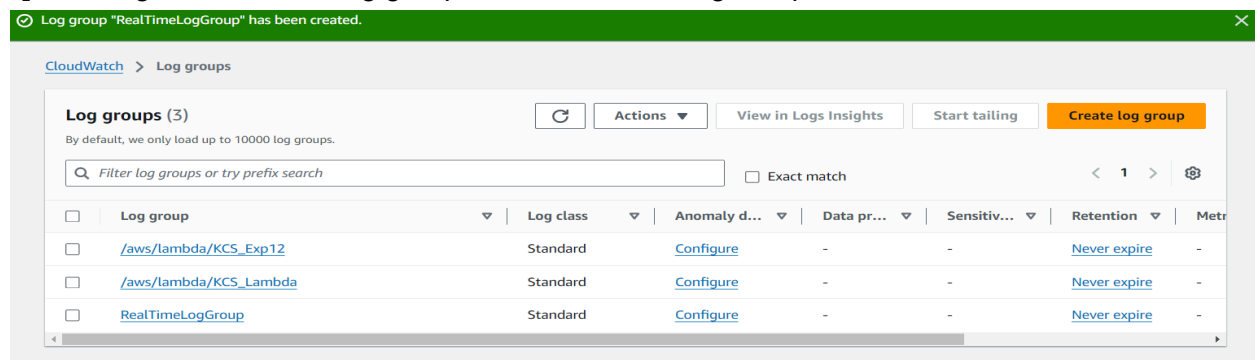
Real-Time Log Processing

MAIN PROBLEM STATEMENT:

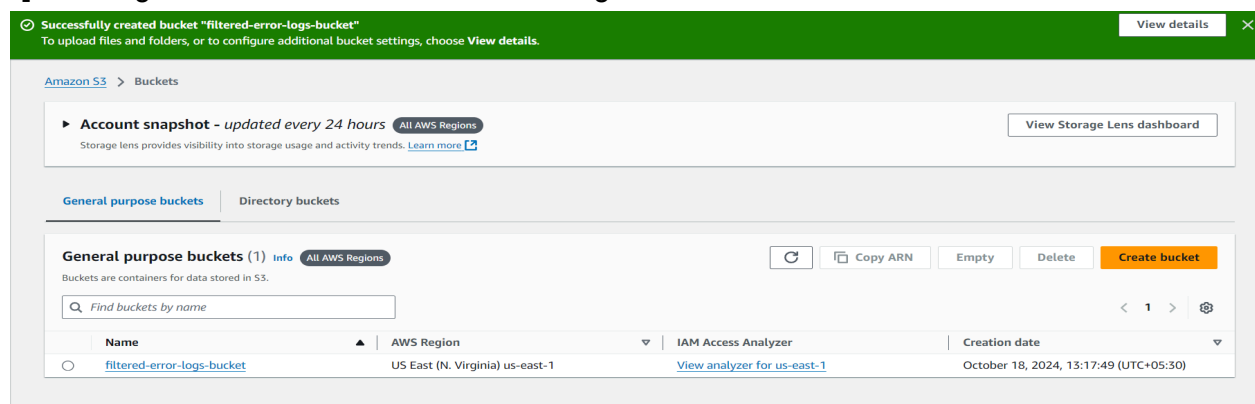
- Concepts Used: AWS Lambda, CloudWatch, S3.
- Problem Statement: "Set up a Lambda function that triggers whenever a new log entry is added to a CloudWatch Log Group. The Lambda function should filter specific log events and store them in an S3 bucket."
- Tasks:
 - Create a CloudWatch Log Group and set up a Lambda function that triggers on new log entries.
 - Write a Python Lambda function to filter logs based on a keyword (e.g., 'ERROR').
 - Store the filtered logs in an S3 bucket.
 - Test by generating logs and checking the S3 bucket for the filtered entries.

STEPS FOR EXECUTION:

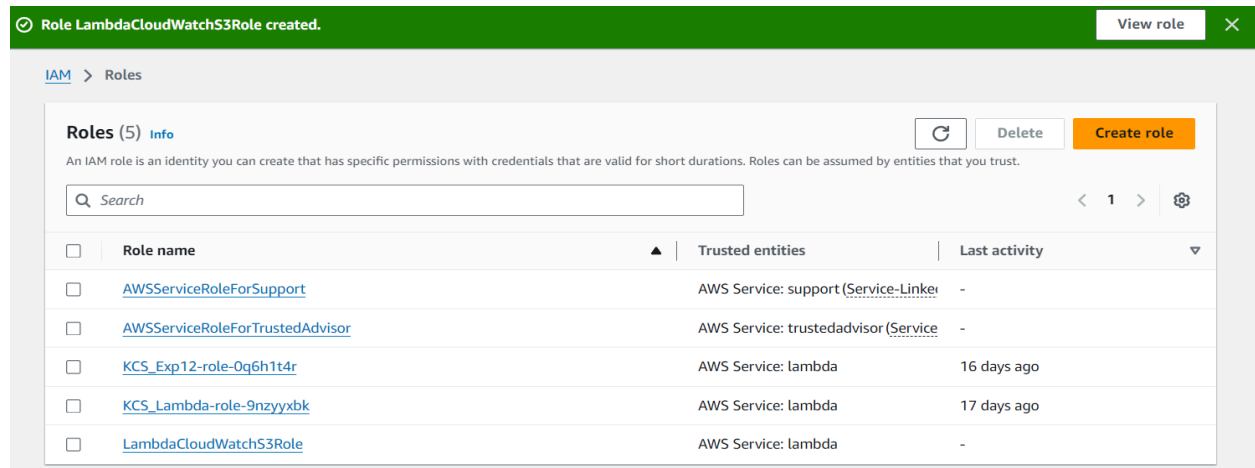
1] Creating a cloudwatch log group called RealTimeLogGroup.



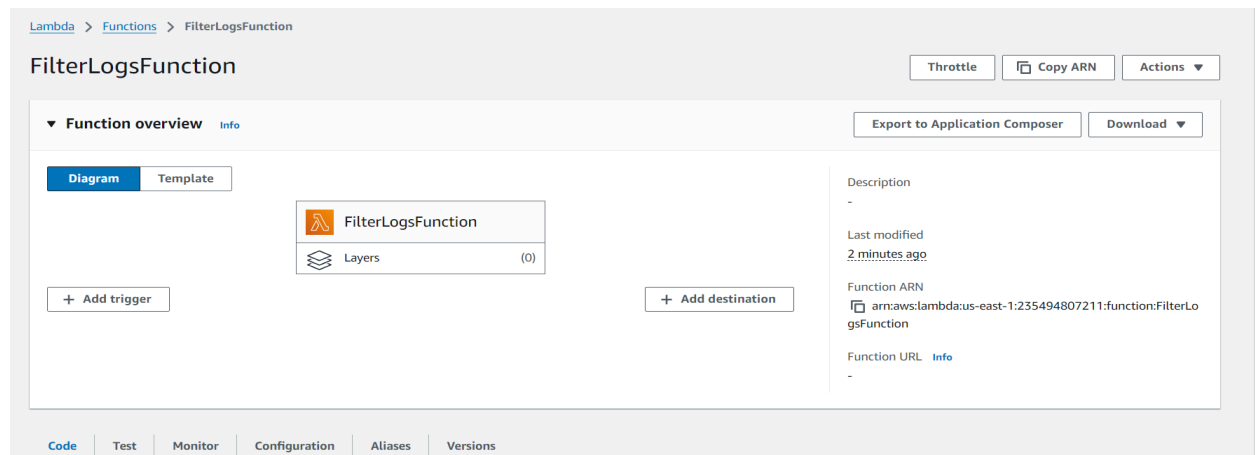
2] Creating a s3 bucket called filtered-error-logs-bucket.



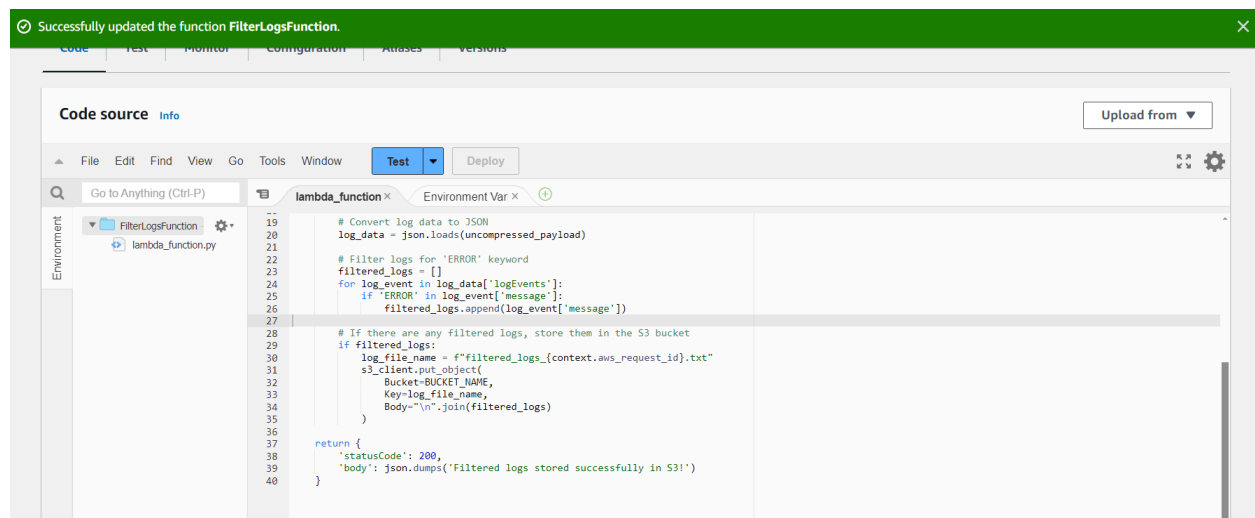
3] Creating an IAM role called LambdaCloudwatchS3Role which grants Lambda permissions to read CloudWatch Logs and write to the S3 bucket.



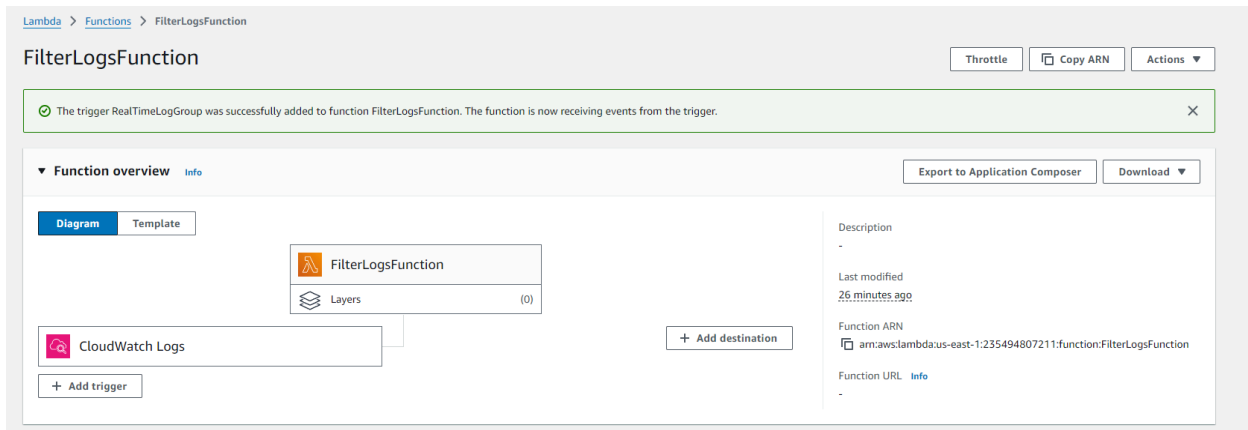
4] Created a lambda function called FilterLogsFunction.



5] Write the following code python code to read log entries from cloudwatch and detect the keyword



6] Create a trigger for the lambda function. It will be triggered when there are entries in our cloudwatch log group.



7] Now we need to generate test logs. There are basically 2 ways to generate test logs. One is using AWS CLI and the other is using AWS cloudwatch console (manual entries.)

I have tried both the ways for pushing log entries.

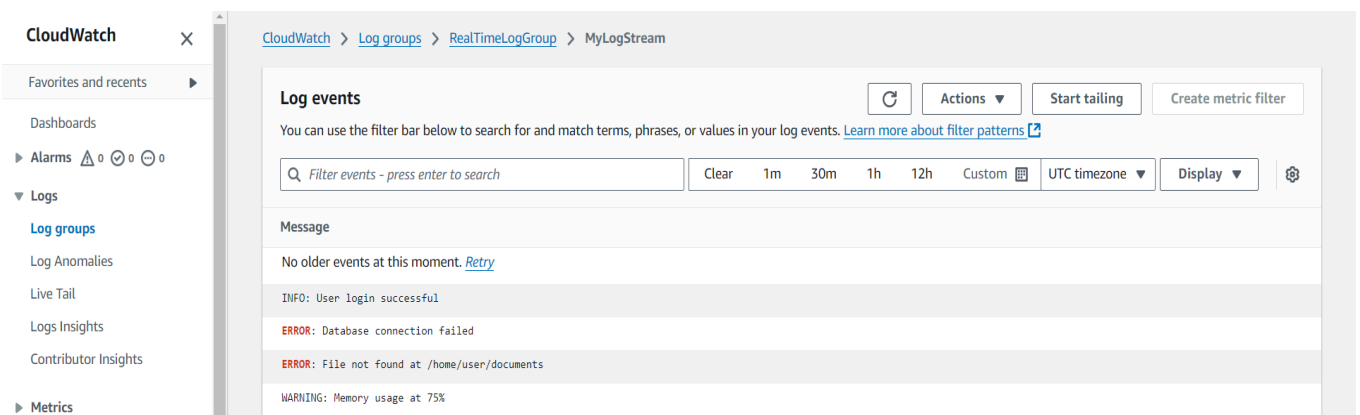
```
Microsoft Windows [Version 10.0.22631.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\91773>aws --version
aws-cli/2.18.9 Python/3.12.6 Windows/11 exe/AMD64

C:\Users\91773>aws configure
AWS Access Key ID [None]: AKIATNVEVS2VYM2SYJR4
AWS Secret Access Key [None]: Fv2ug8+5atBr1lzQImgJusecRYFZ5ogxA2Pj:
Default region name [None]: us-east-1
Default output format [None]:

C:\Users\91773>aws s3 ls
2024-10-18 13:17:49 filtered-error-logs-bucket

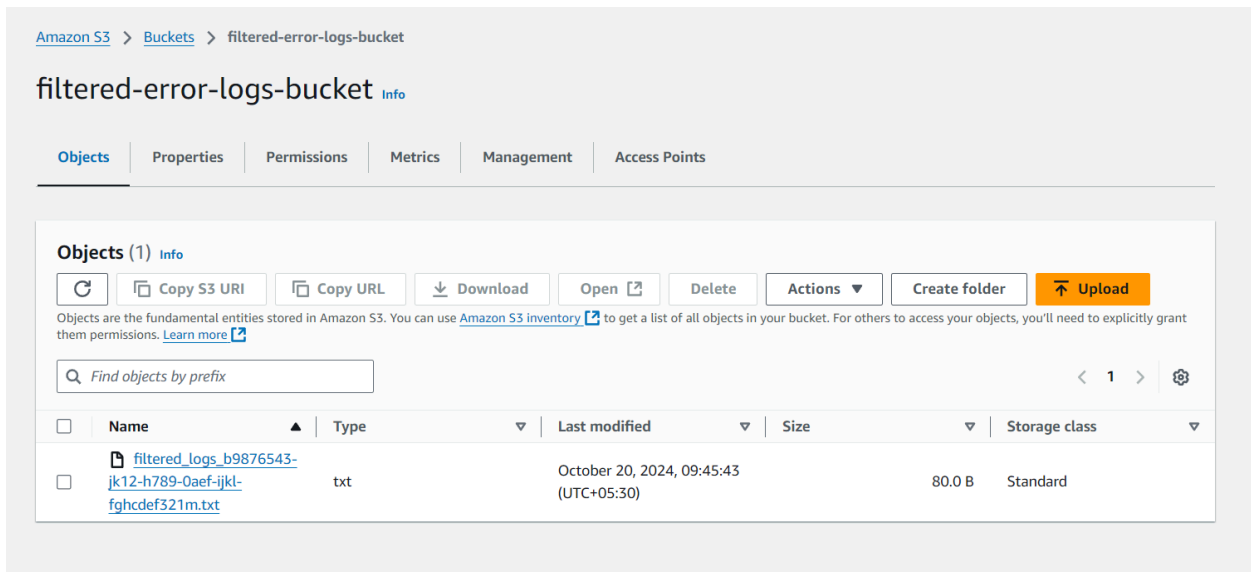
C:\Users\91773>
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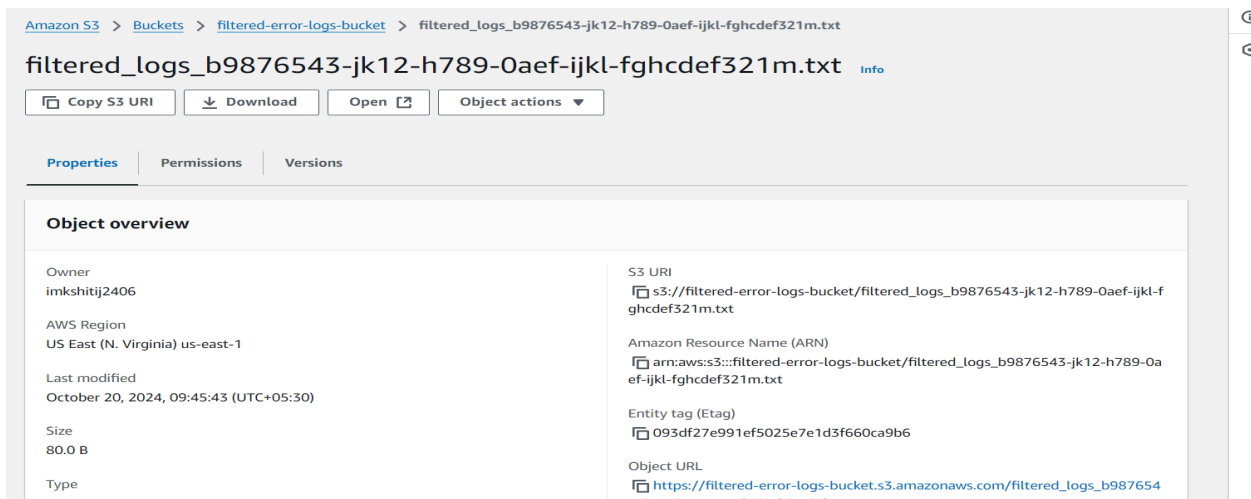
I was facing **multiple errors** while pushing log entries with help from AWS CLI as CLI uses python ver 3.12 which is the latest version of python and is not supported by Lambda. I tackled the error by using python ver 3.9 for lambda.

8] Now since we have generated log entries , our lambda function is triggered. Lambda function actively looks at cloudwatch logs for the Keyword “ERROR”. As soon as it detects the keyword , it stores that specific log in our S3 bucket .

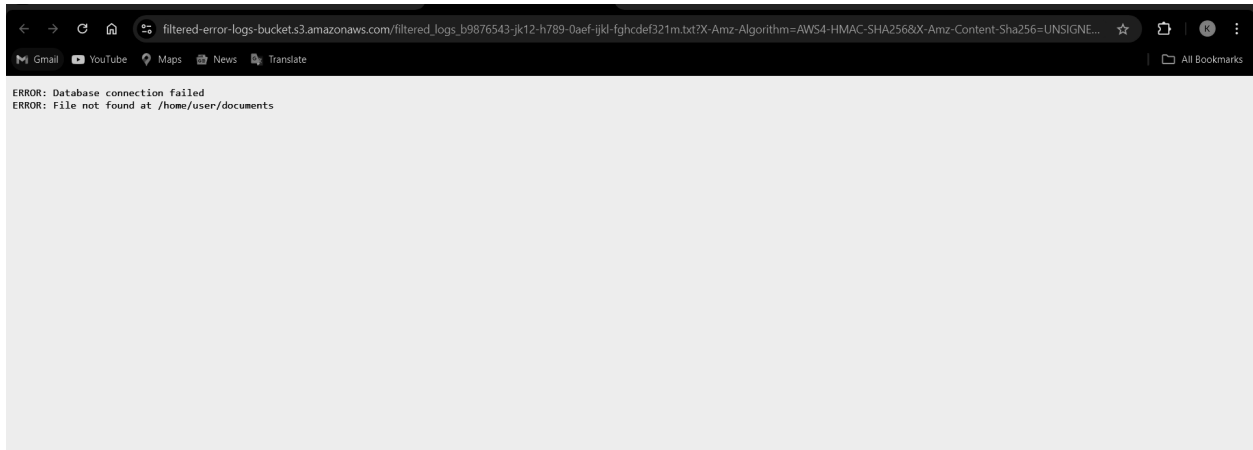
9] To verify the above step, go to the S3 bucket created . You should be able to see a text file .



10] Click on the S3 bucket.



11] Click on Open to see content of S3 bucket. We can see all the errors inside this .txt file which occurred from log entries.



Thus we have successfully performed the case study :)

Conclusion:

I learned how to filter logs in real time, store them in S3, and troubleshoot several issues along the way. Some key challenges included configuring IAM roles, setting up AWS CLI, and ensuring the logs were generated and processed correctly. Despite these hiccups, it was rewarding to see everything come together in the end. This hands-on experience helped me to understand AWS services like Lambda, S3, Cloudwatch and their real life applications. All in all it was a rewarding experience.